

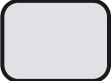
FRI: fold the polynomial in half each round until it is constant

round 0  $\text{deg} < d$

round 1  $\text{deg} < d/2$

round 2  $\text{deg} < d/4$

round 3  ...

 **constant**

log d rounds

prover work $O(d)$

verifier work $O(\log d)$

soundness 2^{12}

Each round is committed by a Merkle root; the verifier checks consistency between consecutive layers at random points.